

HW 8

Townsend Problem 7.22 (5 pts)

Calculate $\langle E \rangle$ and the uncertainty ΔE for the coherent state $|\alpha\rangle$.

To find $\langle E \rangle$, we can do

$$\langle \alpha | \hat{H} | \alpha \rangle$$

Separating this out by components,

$$\begin{aligned} \langle E \rangle &= \frac{\langle p_x^2 \rangle}{2m} + \frac{1}{2} m \omega^2 \langle x^2 \rangle \\ &= \frac{(\Delta p_x)^2 + \langle p_x \rangle^2}{2m} + \frac{1}{2} m \omega^2 [(\Delta x)^2 + \langle x \rangle^2] \end{aligned}$$

eq 7.50 page 257

In an energy eigenstate:

$$\langle E \rangle = \frac{(\Delta p_x)^2}{2m} + \frac{1}{2} m \omega^2 (\Delta x)^2$$

The coherent state $|\alpha\rangle = e^{-\frac{|\alpha|^2}{2}} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$

We know that for the coherent state,

$$\langle x \rangle = \sqrt{\frac{\hbar}{2m\omega}} 2|\alpha| \cos(\omega t + \delta)$$

and that

$$\langle p_x \rangle = -\sqrt{\frac{m\omega\hbar}{2}} 2|\alpha| \sin(\omega t + \delta)$$

$$\begin{aligned} \langle x^2 \rangle &= \frac{\hbar}{2m\omega} [\alpha(t)^2 + \alpha^*(t)^2 + 2|\alpha(t)|^2 + 1] \\ \langle p_x^2 \rangle &= \frac{m\omega\hbar}{2} [2|\alpha(t)|^2 + 1 - \alpha(t)^2 - \alpha^*(t)^2] \end{aligned}$$

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This gives

$$\begin{aligned} (\Delta x)^2 &= \frac{\hbar}{2m\omega} \\ (\Delta p_x)^2 &= \frac{m\omega\hbar}{2} \end{aligned}$$

We can just plug and chug,

$$\begin{aligned} \langle E \rangle &= \frac{(\Delta p_x)^2}{2m} + \frac{1}{2} m \omega^2 (\Delta x)^2 \\ &= \frac{\omega\hbar}{4} + \frac{1}{4} \omega\hbar \\ \langle E \rangle &= \frac{\hbar\omega}{2} \end{aligned}$$

Townsend Problem 9.1 (5 pts)

Follow the suggestion after (9.11) to show that $[\hat{T}(a_x \mathbf{i}), \hat{T}(a_y \mathbf{j})] = 0$ implies $[\hat{p}_x, \hat{p}_y] = 0$.

We have the series expansion

$$\hat{T}(a_x \mathbf{i}) = 1 - \frac{i\hat{p}_x a_x}{\hbar} - \frac{\hat{p}_x^2 a_x^2}{2\hbar^2} + \dots$$

$$\hat{T}(a_y \mathbf{j}) = 1 - \frac{i\hat{p}_y a_y}{\hbar} - \frac{\hat{p}_y^2 a_y^2}{2\hbar^2} + \dots$$

If we apply successive translations in different directions, they don't change anything. We can plug these into

$$\hat{T}(a_y \mathbf{j}) \hat{T}(a_x \mathbf{i}) = \hat{T}(a_x \mathbf{i}) \hat{T}(a_y \mathbf{j})$$

to see that

$$\left(1 - \frac{i\hat{p}_x a_x}{\hbar} - \frac{\hat{p}_x^2 a_x^2}{2\hbar^2} + \dots\right) \left(1 - \frac{i\hat{p}_y a_y}{\hbar} - \frac{\hat{p}_y^2 a_y^2}{2\hbar^2} + \dots\right) = \left(1 - \frac{i\hat{p}_y a_y}{\hbar} - \frac{\hat{p}_y^2 a_y^2}{2\hbar^2} + \dots\right) \left(1 - \frac{i\hat{p}_x a_x}{\hbar} - \frac{\hat{p}_x^2 a_x^2}{2\hbar^2} + \dots\right)$$

We can keep terms up to second order

$$\left(1 - \frac{i\hat{p}_y a_y}{\hbar} - \frac{\hat{p}_y^2 a_y^2}{2\hbar^2} - \frac{i\hat{p}_x a_x}{\hbar} - \frac{\hat{p}_x^2 a_x^2}{2\hbar^2} - \frac{\hat{p}_x a_x \hat{p}_y a_y}{\hbar^2}\right) = \left(1 - \frac{i\hat{p}_y a_y}{\hbar} - \frac{\hat{p}_y^2 a_y^2}{2\hbar^2} - \frac{i\hat{p}_x a_x}{\hbar} - \frac{\hat{p}_x^2 a_x^2}{2\hbar^2} - \frac{\hat{p}_y a_y \hat{p}_x a_x}{\hbar^2}\right)$$

We can see that all the terms are the same except for the very last, so this becomes

$$\hat{p}_x a_x \hat{p}_y a_y = \hat{p}_y a_y \hat{p}_x a_x$$

Dividing by constants and rearranging

$$\hat{p}_x \hat{p}_y = \hat{p}_y \hat{p}_x$$

$$\hat{p}_x \hat{p}_y - \hat{p}_y \hat{p}_x = 0$$

$$[\hat{p}_x, \hat{p}_y] = 0$$

Townsend 9.4 (5 pts)

Use the commutation relations of $\hat{\mathbf{r}}_1$, $\hat{\mathbf{r}}_2$, $\hat{\mathbf{p}}_1$, and $\hat{\mathbf{p}}_2$ to establish that the center-of-mass and relative position and momentum operators satisfy

$$[\hat{x}_i, p_j] = 0 \quad [\hat{x}_i, p_j] = i\hbar\delta_{ij}$$

$$[\hat{x}_i, p_j] = i\hbar\delta_{ij} \quad [\hat{x}_i, p_j] = 0$$

where

$$\hat{\mathbf{r}} = \hat{\mathbf{r}}_1 - \hat{\mathbf{r}}_2 \quad \hat{\mathbf{p}} = \frac{m_2 \hat{\mathbf{p}}_1 - m_1 \hat{\mathbf{p}}_2}{m_1 + m_2}$$

$$\hat{\mathbf{R}} = \frac{m_1 \hat{\mathbf{r}}_1 + m_2 \hat{\mathbf{r}}_2}{m_1 + m_2} \quad \hat{\mathbf{P}} = \hat{\mathbf{p}}_1 + \hat{\mathbf{p}}_2$$

TODO I HAVE NOT FIGURED IT OUT

We have the position momentum commutation relations for individual particles

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

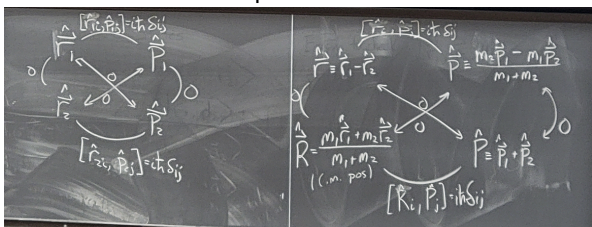
(eq 9.19)

and that generators commute

$$[\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2] = 0 \text{ (eq 9.30)}$$

$$[\hat{r}_{ni}, \hat{p}_{nj}] = i\hbar\delta_{ij}$$

We also have the multiparticle commutations



Townsend Problem 9.5 (5 pts)

Show explicitly that

$$\frac{\hat{\mathbf{p}}_1^2}{2m_1} + \frac{\hat{\mathbf{p}}_2^2}{2m_2} = \frac{\hat{\mathbf{P}}^2}{2M} + \frac{\hat{\mathbf{p}}^2}{2\mu}$$

where

$$\hat{\mathbf{p}} = \frac{m_2 \hat{\mathbf{p}}_1 - m_1 \hat{\mathbf{p}}_2}{m_1 + m_2} \quad \hat{\mathbf{P}} = \hat{\mathbf{p}}_1 + \hat{\mathbf{p}}_2 \quad M = m_1 + m_2 \quad \text{and} \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

$$\frac{\hat{p}_1^2}{2m_1} + \frac{\hat{p}_2^2}{2m_2} \stackrel{?}{=} \frac{\hat{P}^2}{2(m_1 + m_2)} + \frac{\hat{p}^2}{2\left(\frac{m_1 m_2}{m_1 + m_2}\right)}$$

We can bring the left side to look more like this

$$\frac{m_2 \hat{p}_1^2 + m_1 \hat{p}_2^2}{2(m_1 m_2)} \stackrel{?}{=} \frac{(\hat{p}_1 + \hat{p}_2)^2}{2(m_1 + m_2)} + \frac{\left(\frac{m_2 \hat{p}_1 - m_1 \hat{p}_2}{m_1 + m_2}\right)^2 (m_1 + m_2)}{2(m_1 m_2)}$$

I will simplify the right side now

$$\frac{m_2 \hat{p}_1^2 + m_1 \hat{p}_2^2}{2(m_1 m_2)} \stackrel{?}{=} \frac{\hat{p}_1^2 + \hat{p}_2^2 + 2\hat{p}_1 \hat{p}_2}{2(m_1 + m_2)} + \frac{(m_2 \hat{p}_1 - m_1 \hat{p}_2)^2}{m_1 + m_2} \frac{1}{2m_1 m_2}$$

We remove the denominator

$$m_2 \hat{p}_1^2 + m_1 \hat{p}_2^2 \stackrel{?}{=} \frac{m_1 m_2 (\hat{p}_1^2 + \hat{p}_2^2 + 2\hat{p}_1 \hat{p}_2)}{(m_1 + m_2)} + \frac{(m_2 \hat{p}_1 - m_1 \hat{p}_2)^2}{m_1 + m_2}$$

We now get

$$((m_1 m_2 + m_2^2) \hat{p}_1^2 + (m_1^2 + m_1 m_2) \hat{p}_2^2) \stackrel{?}{=} m_1 m_2 (\hat{p}_1^2 + \hat{p}_2^2 + 2\hat{p}_1 \hat{p}_2) + (m_2^2 \hat{p}_1^2 + m_1^2 \hat{p}_2^2 - 2m_1 m_2 \hat{p}_1 \hat{p}_2)$$

The cross term cancels

$$((m_1 m_2 + m_2^2) \hat{p}_1^2 + (m_1^2 + m_1 m_2) \hat{p}_2^2) \stackrel{?}{=} m_1 m_2 (\hat{p}_1^2 + \hat{p}_2^2) + (m_2^2 \hat{p}_1^2 + m_1^2 \hat{p}_2^2)$$

We can see this is true

$$\frac{((m_1 m_2 + m_2^2) \hat{p}_1^2 + (m_1^2 + m_1 m_2) \hat{p}_2^2)}{1} \stackrel{?}{=} \frac{((m_1 m_2 + m_2^2) \hat{p}_1^2 + (m_1^2 + m_1 m_2) \hat{p}_2^2)}{1} = 1$$

So this holds true!